Formal and semantic cues at play in gender assignment to novel words in Italian

Valeria Galimberti[1], Beatrice Giustolisi[1], Caterina Donati[2], Francesca Foppolo[1]

[1]University of Milano-Bicocca, [2]Université Paris Cité

francesca.foppolo@unimib.it

Italian is a highly gender transparent language in which almost all nouns end with a vowel and are grammatically gender-marked. Typically, nouns ending in -a are grammatically feminine (F), those ending in -o are masculine (M). Nouns ending in -e are opaque and can be M or F; nouns ending in consonant (-c) tend to be borrowings, typically anglicisms. Corpus research shows that borrowings tend to be assigned either the gender of the Italian equivalent (semantic analogy) or 'default' M, although some degree of variation is attested [1]. Experimental studies with novel words in Spanish, which has a grammatical system similar to Italian but more frequent consonant endings, reveal a preference for morpho-phonological (over semantic) cues by native speakers and Spanish-dominant early bilinguals [2]. **Our study.** *Which gender is assigned to novel (loan) words in Italian, and what is the contribution of formal and semantic cues?* To address this RQ, we investigate the role of morpho-phonological traits and semantic cues in gender assignment by L1 Italian adults during first exposure to novel words, manipulating word ending (to test the *formal* strategy) and the gender of their translation equivalents (to test the *semantic* strategy). We ran a production study in E-Prime 3 in which a set of images of inanimate objects (N = 180, from the MultiPic database [3]; 50% F in Italian) and pseudo-objects (N = 90, NOUN Database [4]), were associated with pseudowords ending in -a/-o/-c (N = 270), Fig. 1. The picture-pseudoword association was random to control for semantic associations [5], and fully counterbalanced across lists in 3x3 conditions: congruent, incongruent, or opaque, considering the novel *word ending* (made from the same root, e.g., (dimab/dimab-a/dimab-o) and the gender of the *object* used as translation equivalent, including F, M and N (non)-objects. Adult participants (N = 45) were asked to produce a sentence starting with 'è un/una…' (it's $a_{F/M}$) + pseudoword. We coded the gender assigned to the word (M/F) on the basis of the article produced (*un/uno*$_{[M]}$, *una*$_{[F]}$). We also tested children (N = 33, M_{age} = 9:8), and L2 adult learners (in progress). **Results.** *Glmer* (Table 1) on the article produced by adults (Fig. 2-left) show: a prevalence of the formal strategy for words ending in -a (assigned F) and -o (assigned M), and default M for words ending in consonant (-c). Semantic effects emerged in association with F objects: words ending in -c were more likely assigned F compared to words ending in -o. Preliminary results (Fig. 2, right) on the same task administered to children show a higher frequency of the default M strategy compared to adults. **Conclusions.** In line with previous research, we show evidence for a morpho-orthographic strategy in gender assignment and a default-M strategy for novel words ending in -c in Italian. In addition, we find limited but consistent evidence for an influence of the gender of the associated object, at least in L1 adults.

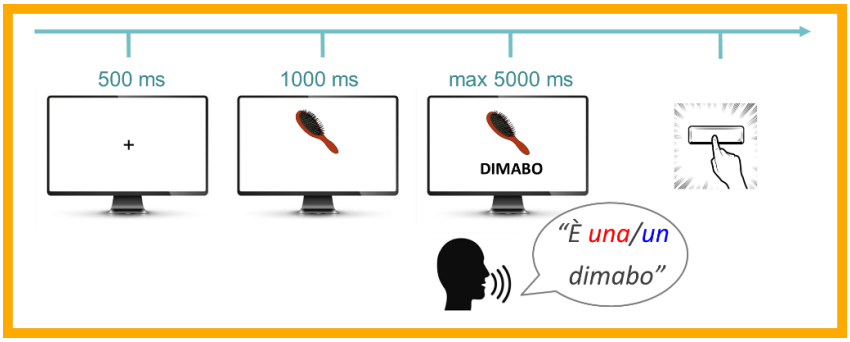

Figure 1. Example of a gender-incongruent experimental trial (F object ('la spazzola', -o ending of novel word 'dimabo').

Table 1. Output of *glmer* on gender assigned (M/F), with Ending (-c, -a, -o) and Object (N/F/M) contrasts set at (0.5, -0.5, 0) and (0.5, 0, -0.5). Formula: gender_assigned ~ Ending*Object + Trial_order + (1|Participant) + (1|Picture) + (1|Word root)

	Est.	S.E.	z val.	p
(Intercept)	-1.537	0.112	-13.731	0.000
Ending_contrast1 (-c vs -a)	-7.068	0.127	-55.689	0.000
Ending_contrast2 (-c vs -o)	5.114	0.195	26.263	0.000
Object_contrast1 (N vs F)	-1.911	0.128	-14.942	0.000
Object_contrast2 (N vs M)	1.051	0.163	6.454	0.000
old20_cent	-0.291	0.126	-2.307	0.021
Ending_contrast1:Object_contrast1	**-1.770**	**0.304**	**-5.826**	**0.000**
Ending_contrast2:Object_contrast1	**1.990**	**0.430**	**4.624**	**0.000**
Ending_contrast1:Object_contrast2	**0.774**	**0.354**	**2.187**	**0.029**
Ending_contrast2:Object_contrast2	-0.321	0.579	-0.554	0.580

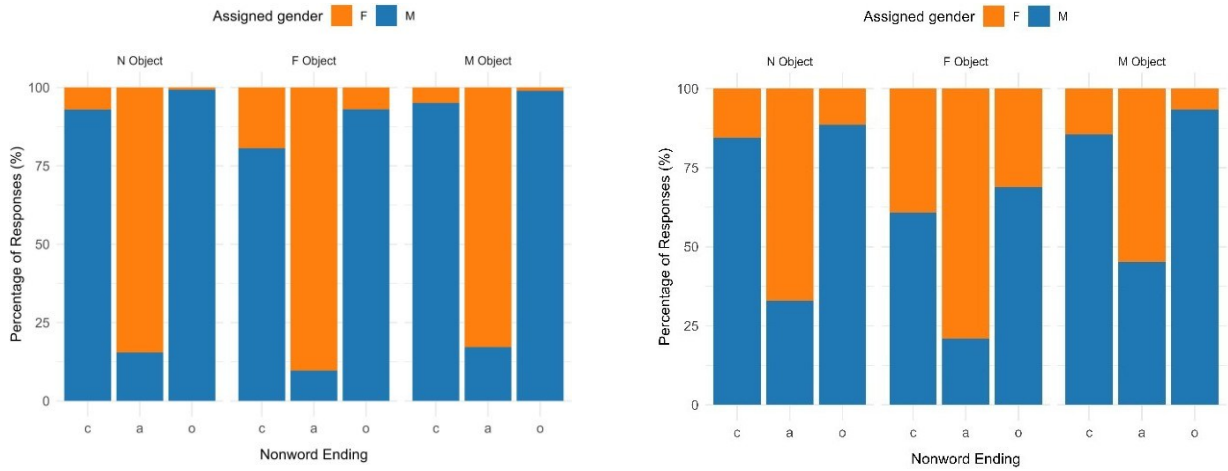

Figure 2. Gender assigned to articles produced by Adults (left) and Children (right)

References [1] Pulcini (2023) *The Influence of English on Italian: Lexical and Cultural Features* De Gruyter [2] Pérez-Tattam et al. (2019) Gender Assignment to Spanish Pseudowords by Monolingual and Basque-Spanish Bilingual Children *Languages* [3] Duñabeitia et al. (2018) MultiPic: A standardized set of 750 drawings with norms for six European languages. *QJEP* [4] Horst et al. (2016). The Novel Object and Unusual Name (NOUN) Database: A collection of novel images for use in experimental research. *BRM* [5] Gatti et al. (2023). Out-of-vocabulary but not meaningless: Evidence for semantic- priming effects in pseudoword processing. *JEP:G*.